10 White dwarfs and neutron stars

A readable and engaging account of the early thermal history of the Universe is in chapter 9 of this ebook: *Statistical mechanics*, R.K. Pathria, Paul D. Beale, Third edition, Elsevier (2011).

Stars of the Sun's mass cannot fuse nuclei of Carbon's charge or more. Once they have reached that state and are radiatively cooling, what supports the star against gravitational collapse? Initially radiation pressure, but eventually degeneracy pressure from the electrons within the star.

The following table (courtesy of Baierlein, p 194) provides data on two of the nearest white dwarfs[1] to Earth, with 40 Eri B being the first observed in 1910.

Star \ Properties	$M/M_\odot$	$R/R_\odot$	$(V/N)^{1/3}$ (10^{-12} m)	$\varepsilon_{\rm F}$ (10^4 eV)	$T_{\rm F}$ (10^9 K)	$p_{\rm F}/(mc)$	$v_{\rm F}/c$	Distance (Light Y)
40 Eri B	0.467	0.013	2.3	6.9	0.8	0.52	0.46	17
Sirius B	1.05	0.0073	0.96	40	4.6	1.2	0.78	8.6

Notice that the mass of Sirius B is greater than 40 Eri B, but is smaller.

10.1 Stability of a non-relativistic star

The two main contributions to the energy of such a star (and hence determine equilibrium) are the gravitational potential energy and the kinetic energy of the electrons. The nuclei are less important, partly because the number density, n, of the electrons is higher by a factor of the average charge on the nuclei. But, more importantly, since the $T = 0$ pressure is $p = \frac{2}{5}n\varepsilon_{\rm F}$ (p 14 of section 9), and the fermi energy of the electrons is $10^4\times$ that of the nuclei, because of the ratio of the masses, the electron contribution to the pressure is several orders of magnitude higher.

An accurate treatment of the equilibrium between gravitational attraction and degeneracy pressure is very difficult, so here we will content ourselves with a highly approaximate treat-

[1] Dwarfs, roofs and oafs are in the minority in terms of plurals formed from words ending in -f. Most become -ves, not -fs, and dwarf (because of Tolkien – who admitted he had made a mistake) and roof are heading in that direction; oaf is stubbornly hanging on.

ment.

We will, unrealistically, assume the star, of radius R, is uniform in composition. The we have:

1. Gravitational potential energy:

$$\mathsf{PE} = -\frac{3}{5}\frac{GM^2}{R} \ .$$

2. To determine the kinetic energy of the electrons, we need to relate the electron density to the mass of the star, which principally comes from the nuclei. If thetotal number of nucleons (protons and neutrons) is A and the total number of protons is $Z = N$, where N is the number of electrons, then there are (A/Z) nucleons per electron. We now neglect the contribution of the electrons' mass to the total mass of the star and moreover neglect the disparity of the proton and neutron mass. Then assuming locally the number of positive and negative charges are equal, then the number density of electrons, n, and the mass density, ρ, may be expressed (m_p is the mass of the proton, and m is the mass of the electron):

$$\rho = \frac{A}{Z} n m_\mathsf{p} \quad \Rightarrow \quad n = \frac{Z}{A m_\mathsf{p}} \rho \ .$$

Then the internal energy, U, due to the electron kinetic energy, may be related to the total mass, M and radius of star, R. We find:

$$\begin{aligned}
U = \frac{3}{5} N \varepsilon_\mathrm{F} &= \frac{3}{5} Z (3\pi^2)^{2/3} \frac{\hbar^2 n^{2/3}}{2m} \\
&= \frac{3}{5}(3\pi^2)^{2/3} \underbrace{A m_\mathsf{p}}_{=M} \frac{Z}{A m_\mathsf{p}} \left(\frac{Z}{A m_\mathsf{p}}\right)^{2/3} \frac{\hbar^2 \rho^{2/3}}{2m} \\
&= \frac{3}{5}(3\pi^2)^{2/3} \left(\frac{Z}{A m_\mathsf{p}}\right)^{5/3} \frac{\hbar^2}{2m} M \left(\frac{M}{R^3}\right)^{2/3} \\
&= \frac{3}{5}(3\pi^2)^{2/3} \left(\frac{Z}{A m_\mathsf{p}}\right)^{5/3} \frac{\hbar^2}{2m} \frac{M^{5/3}}{R^2} \\
&= \lambda \frac{M^{5/3}}{R^2} \ .
\end{aligned}$$

Thus adding the internal energy to the gravitational potential energy we find:

$$\begin{aligned}
E^\mathsf{tot}(M,R) &= -\frac{3}{5}\frac{GM^2}{R} + \lambda \frac{M^{5/3}}{R^2} \\
\Rightarrow \quad \left.\frac{\partial E^\mathsf{tot}(M,R)}{\partial R}\right|_{R=R_*} &= 0 = \frac{3}{5}\frac{GM^2}{R_*^2} - 2\lambda \frac{M^{5/3}}{R_*^3} \\
\Rightarrow \quad R_* &= \frac{5}{3} 2 \frac{\lambda M^{5/3}}{GM^2} \\
&= \frac{10\lambda}{3G} \frac{1}{M^{1/3}} \ .
\end{aligned} \tag{10.1}$$

So the radius decreases with increasing mass. This is consistent with the white dwarf data in the table.

10.2 Stability of relativistic stars

Consider increasing the total mass of the star from the last section. The radius and hence the volume diminishes, so the kinetic energy of the electrons increases. Eventually the electrons become relativistic, and this has a dramatic consequence.

Consider the relativistic expression for the energy of an electron:

$$\frac{\varepsilon_{\mathrm{F}}}{\mathrm{c}} = \sqrt{p_{\mathrm{F}}^2 + m_{\mathsf{e}}^2 \mathrm{c}^2} \ ,$$

so the electrons at the fermi surface are relativistic if $p_{\mathrm{F}} \gtrsim m_{\mathsf{e}}\mathrm{c}$. Let us examine the ultra-relativistic case, where $p_{\mathrm{F}} \gg m_{\mathsf{e}}\mathrm{c}$, it is easier to analyse and it is in that regime where the surprise occurs.

In the ultrarelativistic case,

$$\varepsilon_{\mathbf{k}} = \hbar \mathrm{c}|\mathbf{k}| \quad \text{and} \quad k_{\mathrm{F}} \sim \left(\frac{N}{V}\right)^{1/3} = n^{1/3} \ ,$$

where the second expression is expected on dimensional grounds.

To obtain the internal energy, U, we need to determine the density of states in the relativistic case. The starting point in k-space is as before, but the relation between k and ε is different,

$$\varepsilon = \hbar c k \quad \Rightarrow \quad \mathrm{d}k = \frac{\mathrm{d}\varepsilon}{\hbar \mathrm{c}} \ .$$

So:

$$(2S+1)\left(\frac{L}{2\pi}\right)^d S_d \int_0^\infty \mathrm{d}k\, k^{d-1} = (2S+1)\left(\frac{L}{2\pi}\right)^d \frac{S_d}{(\hbar \mathrm{c})^d} \int_0^\infty \mathrm{d}\varepsilon\, \varepsilon^{d-1}$$

$$\Rightarrow \quad g^{\mathsf{rel}}(\varepsilon) = (2S+1)\left(\frac{L}{\mathsf{hc}}\right)^d S_d\, \varepsilon^{d-1} \ . \tag{10.2}$$

Now we determine ε_{F} as before, by ensuring there are N electrons in the system. Thus

$$\begin{aligned} N &= \int_0^{\varepsilon_{\mathrm{F}}} \mathrm{d}\varepsilon\, g^{\mathsf{rel}}(\varepsilon) \\ &= (2S+1)\left(\frac{L}{\mathsf{hc}}\right)^d S_d\, \frac{{\varepsilon_{\mathrm{F}}}^d}{d} \end{aligned}$$

$$\Rightarrow \quad \varepsilon_{\mathrm{F}}(d) = \left(\frac{dn}{2S+1}\,\frac{1}{S_d}\right)^{1/d} \mathsf{hc} \ . \tag{10.3}$$

In 3d we find explicitly:

$$\varepsilon_{\mathrm{F}}(3) = \left(\frac{3}{8\pi}\, n\right)^{1/3} \mathsf{hc}\ .$$

We determine U in the same manner as the nonrelativistic case:

$$\begin{aligned} U &= \int_0^{\varepsilon_{\mathrm{F}}} \mathrm{d}\varepsilon\, \varepsilon g^{\mathsf{rel}}(\varepsilon) \\ &= (2S+1)\left(\frac{L}{\mathsf{hc}}\right)^d S_d \frac{{\varepsilon_{\mathrm{F}}}^{d+1}}{d+1} \\ &= \underbrace{(2S+1)\left(\frac{L}{\mathsf{hc}}\right)^d S_d\, \frac{{\varepsilon_{\mathrm{F}}}^{d}}{d}}_{=N} \left(\frac{d}{d+1}\right) \varepsilon_{\mathrm{F}} \\ \Rightarrow \quad U &= \left(\frac{d}{d+1}\right) N\varepsilon_{\mathrm{F}}\ . \end{aligned} \tag{10.4}$$

This allows us to deduce the degeneracy pressure as in the nonrelativistic case:

$$\begin{aligned} p = -\frac{\partial U}{\partial V} &= -\left(\frac{dN}{(2S+1)S_d}\right)^{1/d} \mathsf{hc}\frac{\partial}{\partial V}\frac{1}{V^{1/d}} \\ &= \frac{1}{d}\frac{U}{V}\ . \end{aligned} \tag{10.5}$$

We may now write the internal energy, U, in terms of the radius, R, and mass, M, of the star in three dimensions, noting $n = N/V$,

$$\begin{aligned} U &= \frac{3}{4} N \left(\frac{3n}{8\pi}\right)^{1/3} \mathsf{hc} \\ &= \frac{3}{4}\left(\frac{3}{8\pi}\right)^{1/3} \mathsf{hc}\frac{(M/m_{\mathsf{p}})^{4/3}}{R} \\ &= \widetilde{\Lambda}\frac{M^{4/3}}{R}\ . \end{aligned} \tag{10.6}$$

If we form the total energy now,

$$E^{\mathsf{tot}} = -\frac{3}{5}\frac{GM^2}{R} + \widetilde{\Lambda}\frac{M^{4/3}}{R}\ ,$$

there is no equilibrium, as no stationary point as a function of R (both terms proportional to $1/R$). Either the star gravitationally collapses if

$$M^{1/3} > \frac{5\widetilde{\Lambda}}{3G}\ ,$$

or the star expands till it becomes nonrelativistic if the mass is less than this, *Chandrasekhar limit*.

If it collapses it may become a neutron star, supported by degeneracy pressure of the neutrons, not the electrons, or a black hole.